2.3 Protect your computer

In this lesson

You will learn:

- how to protect your computer against malware and hackers.



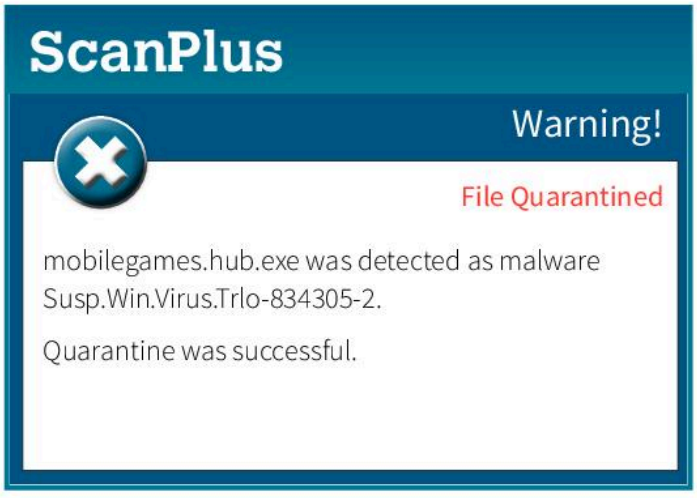
Anti-virus software

You can prevent malware from doing damage by installing **anti-virus software** (**AV software** for short) on your computer. AV software deals with all malware threats, not just viruses. AV software sits on your computer, looking out for malware.

How AV software works

Every piece of malware has a fingerprint. The fingerprint is a piece of programming code that is unique to the malware. This fingerprint is called a **signature**. AV software contains a database of malware signatures.

AV software continually checks the files on your computer, looking for malware signatures. If your AV software finds a malware signature in a file, it puts the file in **quarantine** – this means that the computer cannot open the file.



Modern AV software protects your computer very well. But there may be times when AV software fails to detect malware. If a new piece of malware is developed, your AV software will not recognise it because the signature of the malware will not be in the signature database.

AV software updates the signature database regularly. But there will always be times when your computer is vulnerable to new malware.

Activity

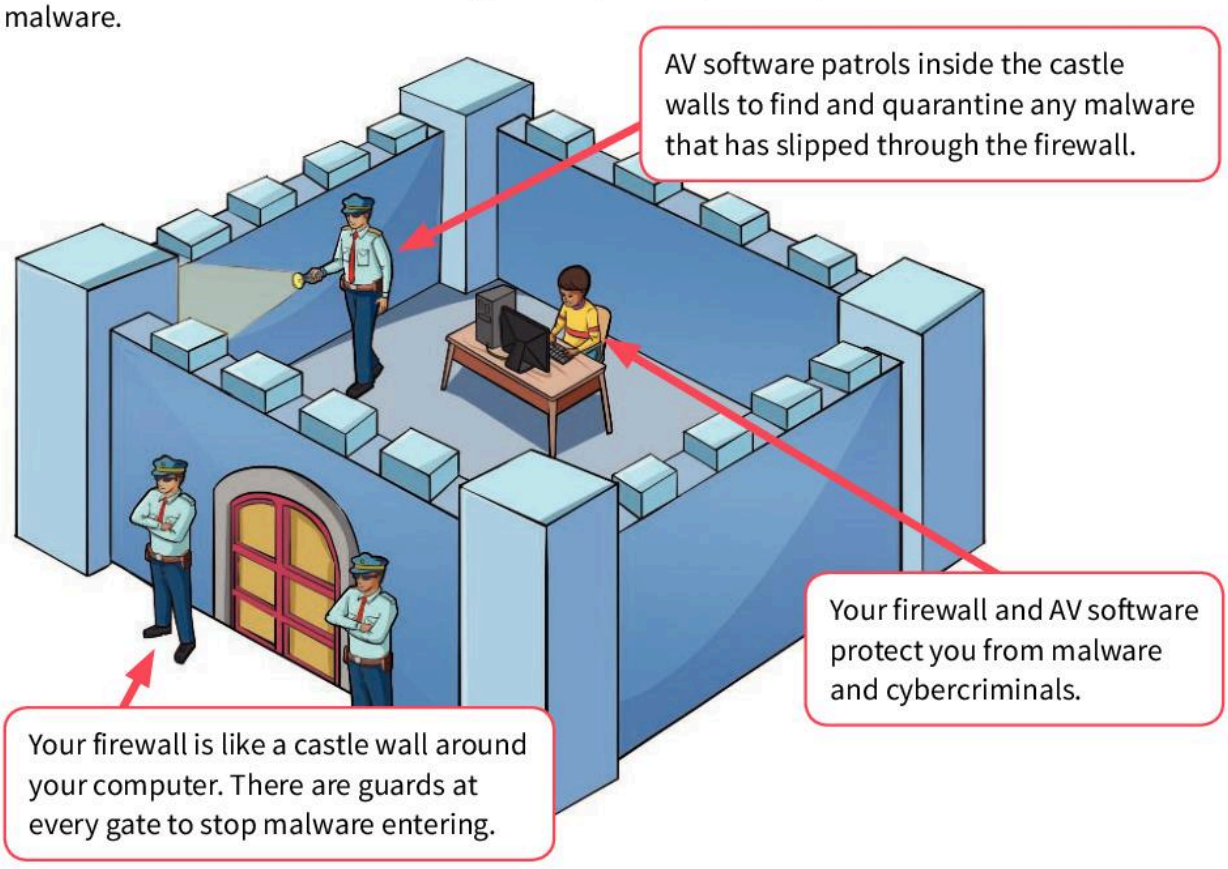
There are many companies that make AV software. Norton and Kaspersky are two examples. Search the web for the names of other AV software packages. Find reviews of different brands of AV software. Make a note of the good and bad points in the reviews for each product.

Firewall

AV software detects and quarantines files that contain malware. Another type of software that protects your computer is a **firewall**.

A firewall surrounds and protects your computer like the walls of a castle. A firewall checks all data before it allows the data to pass through the gates in the wall. Only data that comes from a safe source can pass through the wall.

Your firewall and AV software work together to protect your computer from malware.



Using your computer safely

You can improve your security by behaving responsibly and safely when you use your computer.

Use firewall and AV software

Never turn off the firewall or AV software on your computer. Never change any of the settings in the software. If you do, your computer is open to malware.

Use email safely

Malware can come from files attached to emails. If you open an email attachment, you may be inviting a criminal onto your computer.

Criminals and hackers use misleading email titles to tempt you to open a mail. For example:

"Congratulations! You have won first prize in our free draw."

"Warning! Your account is about to be closed. You MUST act now!"

Be wary of emails from people you do not know. Even emails that look genuine can be clever fakes. An honest email will never ask you to give away personal information or login details.

Spiral back



In Student Book 5 you learned about using strong passwords. Always use strong passwords that are difficult for a hacker to guess.

Use the web safely

When you are searching the web, only use sites that you trust. Files that you download can contain malware. Downloading apps and games is particularly risky. If you are downloading software, always use an official site.

When browsing the web, you will sometimes see messages that tell you there is a problem with your computer. The message will tell you that you need to download software to fix the problem. Do not download the software. Close the web page.

We have detected a problem with the virus protection on your computer.
You should scan for viruses immediately.
Download Virus Scanner

Never click on links in messages that appear when you are online.

Update your software

Keep the software on your computer up to date. Software companies constantly check their software for errors. Sometimes they find security errors. Criminals and hackers use security errors to access computers.

If your software provider sends you an update message, update your software immediately. The updated version will make it more difficult for criminals to install malware. If you think an update message may be malicious, check with an adult before accepting it.

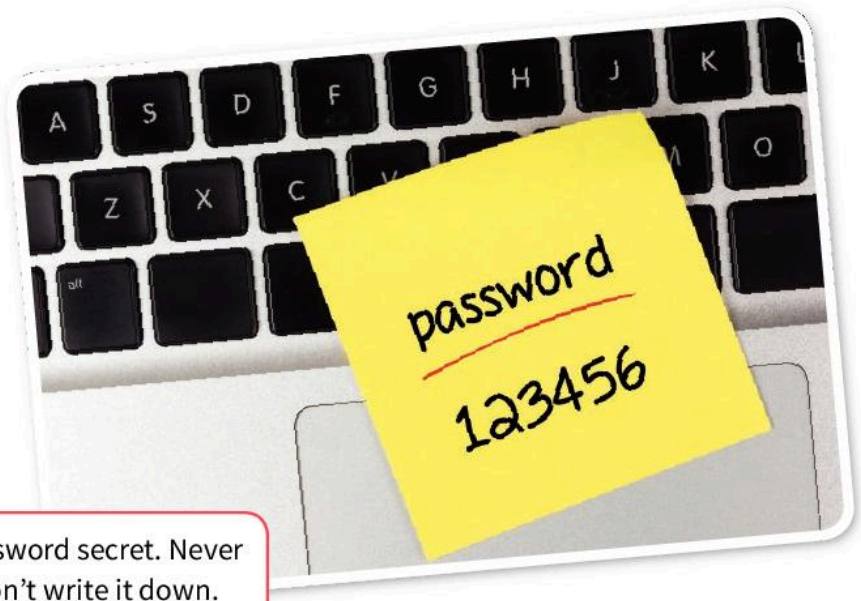


Your software provider will send update messages to a window on your screen, never as an email message.

Keep your password safe

Use a strong password that contains upper and lower case letters, numbers and other characters (for example, $). A strong password should be at least eight characters long

Change your password regularly. That makes it more difficult for hackers to access your computer even if they do manage to steal your password. If you think your password has been stolen, change it immediately.



Keep your password secret. Never share it and don't write it down.

Activity

Write a guide for students on how to work safely online and avoid the risk of malware.

Extra challenge

In Student Book 5 you learned the passphrase method to create strong passwords. Search the web to find another method for creating a strong password. Write down instructions with an example. Add these to your guide on working safely online.

Did you know?

Hackers use special software to crack passwords. A weak password that is 7 characters long can be cracked in less than 1 second. A strong password that is 12 characters long could take 200 years to crack.

Test

1 Name two types of software you can use to protect your computer from malware.
2 Why is it important to update the software on your computer regularly?
3 Explain how AV software and a firewall work together.
4 Explain why you need to be careful when you open a file attached to an email.

2.4 Cyberbullying

In this lesson

You will learn:

- that bullying can be a danger on the internet
- how cyberbullies work
- what to do if you experience cyberbullying.

What is cyberbullying?

A person who uses the internet to bully people is called a **cyberbully**. Cyberbullying is a serious problem. In some countries, over 50% of young people have experienced cyberbullying. Cyberbullying often takes place on social media sites, but bullies also use email and text messages.

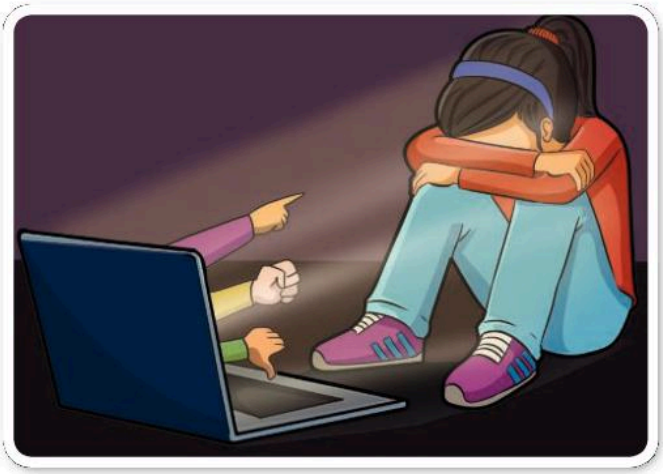
Methods cyberbullies use

- **Harassment** – A bully sends threatening messages by text message or instant messaging. This is a serious form of bullying. The messages often make physical threats. They can be very frightening.
- **Impersonation** – A bully chooses a target (a person they want to bully) and then sends messages that pretend to be from the target. The messages are sent to people who know the target. The aim is to cause trouble between the target and their friends.
- **Exclusion** – A bully sends messages that publicly exclude the target from social groups and events. For example, the bully posts a message on a social media site saying that the target is not invited to an event.
- **Dissing** – A bully circulates rumours (untrue stories and opinions) about a target. The rumours are designed to humiliate and ridicule the target.
- **Photographs** – Some cyberbullies use their smartphone camera to take photos of a target. They then share the photos by text message or social media. They sometimes use the photos in harassment messages.



Be creative

Write a short story about cyberbullying. Choose one of the methods that cyberbullies use. Describe an incident when a bully used that method. How did the target feel about the bullying? What did they do about it? Why did the bully act that way?



The effects of cyberbullying

Cyberbullying is malicious and persistent. It can seem impossible to escape from. A bully can strike at any time, using text messages and social media posts. The internet brings the bullying into the target's home.

Bullying over the internet is often anonymous. Not knowing who is carrying out the bullying makes it more frightening. Cyberbullying can be devastating if you are the target. It can affect every aspect of life.

Confidence

No matter how much a target tries to ignore cyberbullying, the bullying will affect their self-confidence. Low self-confidence can lead to anxiety and depression. Being the constant target of rumour and ridicule makes a person feel powerless and vulnerable.

Loneliness

Being bullied is a lonely experience. A person who is bullied often feels that no-one can help them. They feel isolated, which may cause difficulties in their relationships with family and friends. The target of cyberbullying can withdraw from their family and social groups.



Performance at school

A target of bullying sometimes does less well at school. The cyberbullying distracts them from their schoolwork and they find it difficult to concentrate in class. In extreme cases the person may start to miss classes. They no longer enjoy learning or being with school friends.

Change of personality

Being bullied can change someone's behaviour and personality. A person who is usually pleasant and respectful may start to misbehave or become aggressive. The stress caused by bullying can also lead to problems with sleeping and eating routines. The person may even become ill.

What makes a cyberbully?

If you are the target of a cyberbully, it is easy to think that you are to blame in some way for the bully's actions. It is important to realise that you are not to blame. There are many reasons why someone might become a cyberbully.

The cyberbully wants revenge for harm that has been done to them. It is rarely the person they bully who has caused the harm.

They turn their own unhappiness and insecurity on to someone else.

They want other people to like and admire them.



They are jealous of the person they bully.

Online bullying allows the bully to be anonymous so they believe they can get away with their bullying.

The cyberbully feels threatened by people who are different to themselves.

What should you do if you experience cyberbullying?

If you see someone being bullied

- Be careful not to join in. When you see a post that makes fun of someone, never click the 'like' button – not even as a joke. You will encourage the cyberbully and upset the person who the post is about.
- Say something positive about the person being bullied. Showing your support will make the bullying easier to bear.
- Offer support and friendship. Let the person talk if they need to. Encourage them to talk to an adult about the bullying.



If you are bullied

- Collect evidence. Take screenshots of posts so that you have a record even if the posts are deleted.
- Talk to someone you trust about the bullying. Talk to a friend, family member or teacher.
- Don't respond to the bullying. Responding can encourage the bully and make the situation worse.
- Spend some time offline.



Activity

Write an information sheet on how to survive cyberbullying. If you have the time to do the Extra challenge, use some of the information you discover in your web search.

Extra challenge

Search the internet to find one or more sites that give good information on how to deal with cyberbullying. Add the sites to your bookmark list.

Test

1. List the actions you should take if you are bullied online.
2. Explain two methods that a cyberbully uses.
3. Explain why someone might decide to bully another person online.
4. What should you do if you receive a threatening message on your phone or computer?

